Challenges and Successes in the Energy Transition

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IAE-CSIC & BSE

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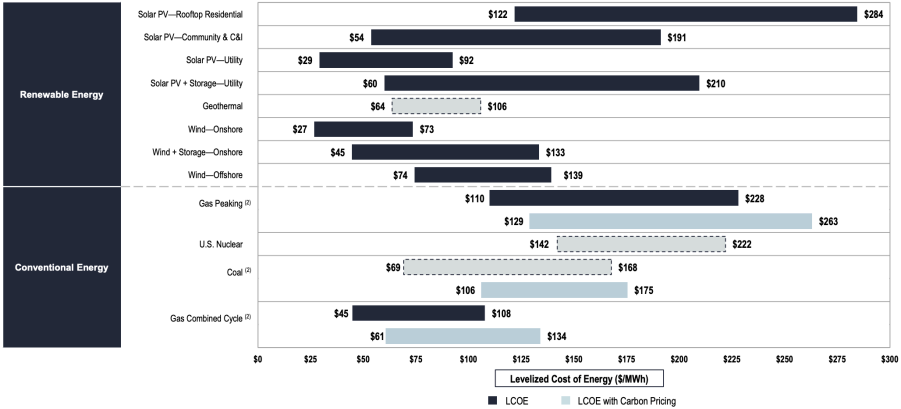
Big push in the electricity sector to decarbonize and electrify

- Need to reduce Green House Gas emissions (GHGs).
- Electricity sector (≈35-40% of CO_2 emissions) has been **most active** and has the greatest potential in making the transition.
- Ambition to move towards **carbon-free electricity/energy** by 2050.
- **Limits to decarbonization**:
  - **Renewables' intermittency** might lead to a potential mismatch between supply and demand, increasing need for flexibility.
  - **Need to improve complementary infrastructure** in high and low voltage.
  - **Growing pressures** due to decarbonization of other sectors (cars, heating, AI, etc.).
  - **Vulnerabilities** due to climate shocks.

Renewables are cost-effective

Levelized Cost of Energy Comparison—Sensitivity to Carbon Pricing

Carbon pricing is one avenue for policymakers to address carbon emissions; a carbon price range of $40 – $60/Ton[1] of carbon would increase the LCOE for certain conventional generation technologies, as indicated below



Source: *Lazard and Roland Berger estimates and publicly available information.*
Note: Unless otherwise noted, the assumptions used in this sensitivity correspond to those used in the LCOE analysis as presented on the page titled "Levelized Cost of Energy Comparison—Version 17.0".
(1) In November 2023, the U.S. Environmental Protection Agency proposed a $204/Ton social cost of carbon.
(2) The low and high ranges reflect the LCOE of selected conventional generation technologies including an illustrative carbon price of $40/Ton and $60/Ton, respectively.

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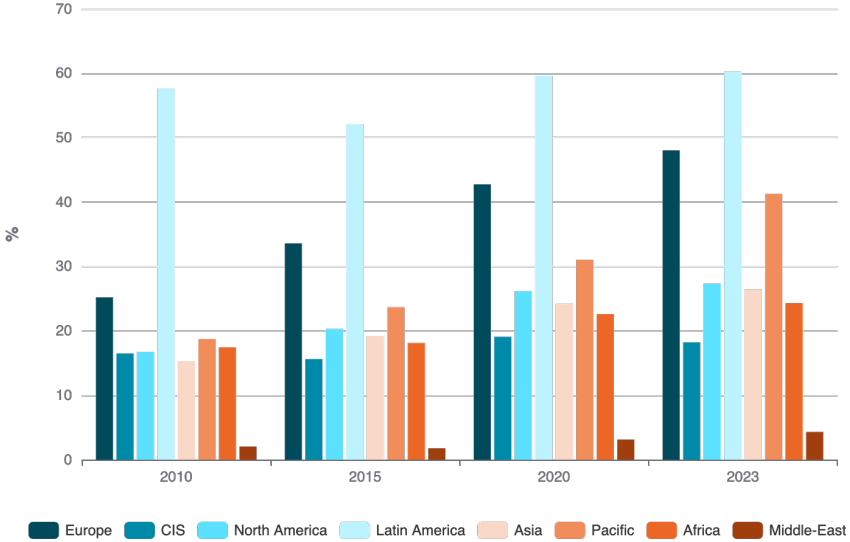
With growing presence, but far from “net-zero” in most regions



Substantial challenges remain...

- Several concerns could hinder the advancement of the energy transition:
  - Intermittency and frequency regulation
  - Transmission and reliability, infrastructure failures
  - Stranded assets and the cost of financing
  - Acceptability and equity, pricing, and job transitions
  - Fiscal pressure even within climate policies: adaptation & mitigation
  - Geopolitical reshaping of trade, e.g., with carbon pricing and new tariffs.
  - Climate risks and the need for resilience.
  - Etc.
- I will talk about three of these issues with examples from my research.

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I. Case study from Spain: Intermittency

II. Case study from Chile: Transmission

Conclusion

I. Case study from Spain: Intermittency

The Impacts of Wind Power in Spain

- **Question:** What have been the impacts of wind generation in the last decade?
- **Methods:** Regression analysis of hourly operational data (prices, congestion costs, emissions benefits, etc.).
- **Co-authors:** Claire Petersen and Lola Segura-Varo
- **Findings:**
  - Consumers have been better off, even after accounting for the cost of the subsidies. Market design can impact these benefits.
  - Innovation in how the market operates has contributed to increase the benefits.

Several studies explore the benefits

- Cullen (2013) and Novan (2015) measure the emissions reductions benefits from wind production.
- Bushnell and Novan (2021) measure the price impacts of solar in California.
- Abrell, Kosch, & Rausch (2019) assess impacts of wind and solar in Germany and Spain.
- Liski, M., & Vehviläinen (2020) assess impacts of wind in Nordic market.
- Gowrisankaran, Reynolds, & Samano (2016) build a structural model to analyze optimal reliability policies.
- Fell, Kaffine, and Novan (2021) look at environmental impacts of renewables with more transmission
- ...

We focus on the **cost of intermittency** in this paper.

Data

- We get hourly data from the Spanish electricity market (2009-2018). Data from REE and OMIE.
- Data include: market prices, intermittency costs, congestion, and other reliability services, emissions data (tons/CO2), subsidies received (millions), etc.
- We **quantify the impact of wind** on these variables:
  - Benefits: emissions reductions, reduced use of fuels, price reductions for consumers.
  - Costs: increased costs of intermittency (paid by consumers and by wind farms), price reductions for consumers.

Identification strategy

- Given randomness in wind forecasts, we run a regression of the impacts of wind on these variables.
- **Spline approach** to look at the impact at different quintiles:

$$Y_t = \beta_0 + \sum_{q=1}^{5} \beta_q W_{qt} + \gamma X_t + \epsilon_t \ ,$$

  where W_{qt} are spline bins according to the quintiles of the wind variable.
- Examine *average* predicted costs as well as *marginal effects*.

Note on endogeneity

- Wind production can be endogeous due to:
  - Curtailment.
  - Strategic behavior.
- Use forecasted wind either directly or as an instrument to actual production.

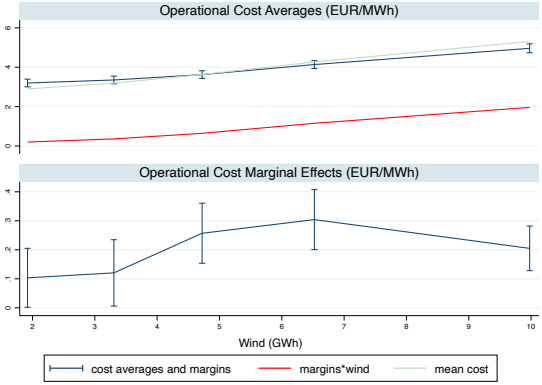
VARIABLES	(1) Wind Forecast	(2) Wind	(3) IV Forecast	(4) IV Power
Forecasted wind (GWh)	0.191			
	(0.0162)			
Final wind production (GWh)		0.152	0.182	0.188
		(0.0140)	(0.0150)	(0.0189)
Observations	83,840	83,841	83,840	81,348
R-squared	0.561	0.557	0.079	0.079

Emphasis on operational costs

- In the literature, often large emphasis on the costs of intermittency from renewable resources.
- Focus on the paper to quantify intermittency costs in the market.
- *Has wind contributed to large increases in operational costs?*
- We identify intermittency costs as the (accounting) costs of providing congestion management, reliability services, balancing, etc.

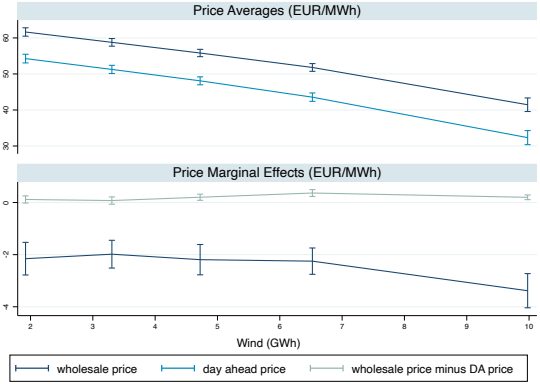
Results for operational costs

- Operational costs go up with more wind.
- However, they don't increase dramatically.
- Marginal effects don't increase.



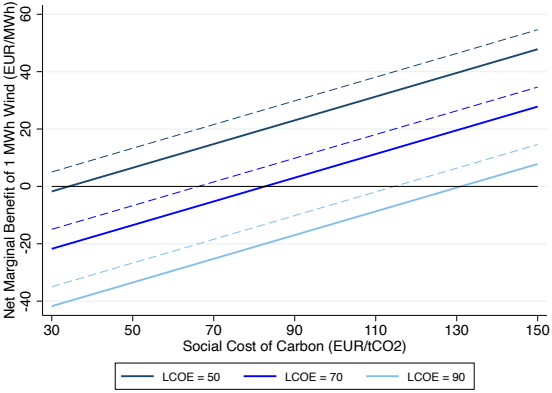
Results for prices

- Wind reduces prices in the market.
- Effect is one order of magnitude larger than the effect on operational costs.



Cost-benefit for different SCC and LCOE

- Overall cost benefit sensitive to assumptions on the cost and benefits of wind power.
- LCOE = (mostly) capital costs of wind.
- SCC = social cost of carbon, global environmental benefits.
- Intermittency has some impacts, but does not affect qualitative findings.



Intermittency and the importance of market design

- There have been discussions on the value of renewables due to their intermittency and the presence of technical constraints, even more after the blackout.
- The costs of integrating wind power into the electricity market can depend on **how well-designed the market is**.
- Market design also interacts with **subsidies**.
  - E.g., negative prices in Texas or Germany, zero prices in Spain.
- Several markets have adapted their functioning to accommodate renewable power:
  - *California:* EIM market to allow for trade between regions.
  - *Germany:* half-hour markets (instead of hourly).
  - *Europe:* move towards continuous trading to have more flexibility.

Summary

- Wind investments had a positive impact on welfare for reasonable SCC.
- On average, the policy benefited both consumers and (wind) producers.
- Details on market design and compensation can substantially impact winners and losers.
- Sometimes perceived as a costly mistake, but a huge early success in climate policy has led to over 20% of generation in Spain being from the wind.
- Regulatory changes can provide useful innovations that reduce costs.

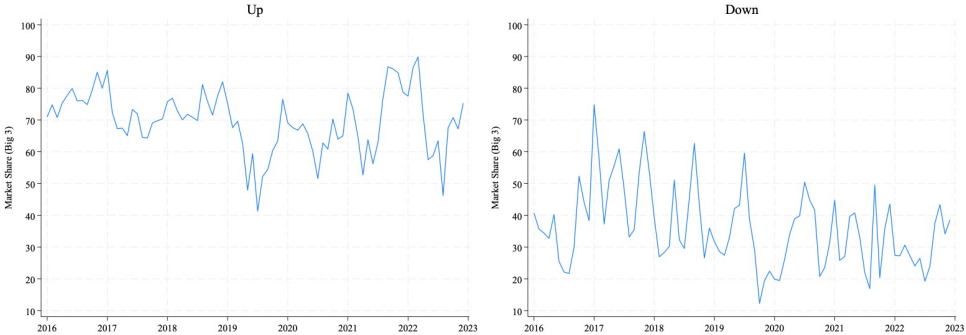
What are the impacts with solar?

Brown and Reguant (2026)

- **Question:** How are reliability markets working and how do they need to be redesigned? How does the impact differ with wind and solar?
- **Methods:** Detailed analysis of bidding data at the unit level, with a combination of descriptive statistics (for now) and structural modeling (TBD).
- **Data details:**
  - Spanish data on reliability markets has an unprecedented level of detail.
  - We can examine the bidding strategies of every single firm, how they update it over time, how it interacts with the day-ahead and other energy markets, etc.

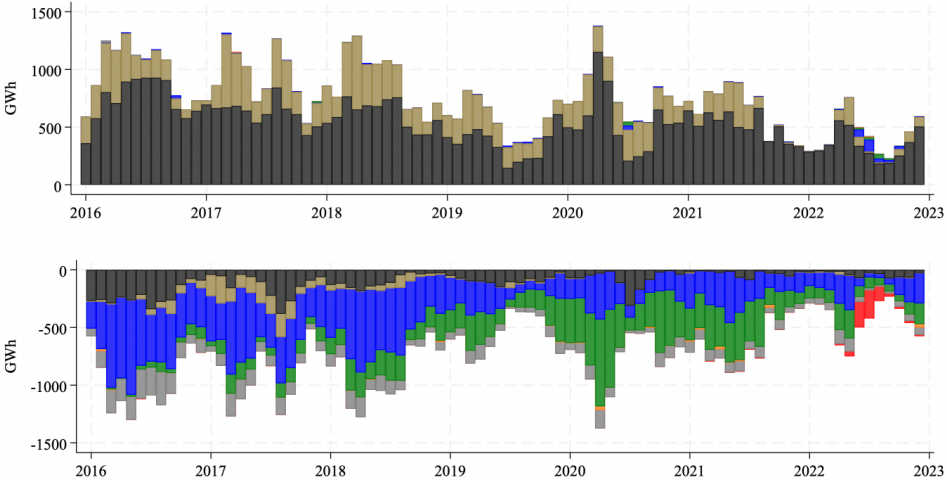
We see evidence of concern

- Back-up power is especially concentrated in the top firms, oftentimes with very few power plants offering their services in a given regulation area.
- This can make the system fragile, as observed during the blackout.



Competition among the Last of Them

- Competition in the restriction market (and related products) will only increase as newer sources (e.g., batteries, hybrid renewable installations) are incorporated in the market.



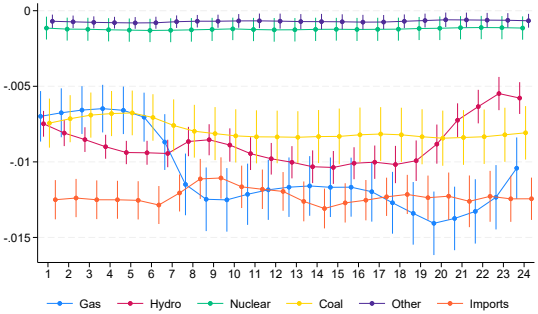
Additional costs from solar larger, but still gains overall

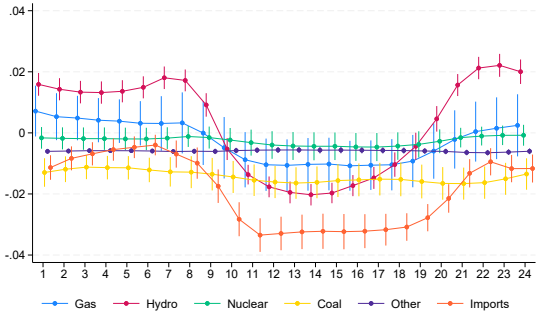
Table 2: Marginal Daily Impacts on Prices by Market Segment

	(1) DA price	(2) AS costs	(3) Final price	(4) DA price	(5) AS costs	(6) Final price
Daily Solar	-0.160	0.052	-0.108	-0.182	0.023	-0.159
	(0.029)	(0.004)	(0.028)	(0.042)	(0.005)	(0.041)
Daily Wind	-0.189	0.018	-0.171	-0.186	0.013	-0.173
	(0.017)	(0.002)	(0.016)	(0.026)	(0.002)	(0.025)
Daily Hydro	-0.160	0.030	-0.129	-0.104	0.020	-0.084
	(0.032)	(0.006)	(0.028)	(0.083)	(0.008)	(0.079)
Daily Demand	0.168	-0.023	0.145	0.149	-0.019	0.130
	(0.016)	(0.002)	(0.014)	(0.026)	(0.003)	(0.025)
MC CCGT	1.040	0.019	1.060	1.012	0.019	1.031
	(0.016)	(0.004)	(0.016)	(0.032)	(0.009)	(0.033)
Observations	3,256	3,256	3,256	3,256	3,256	3,256
R-squared	0.905	0.676	0.917	0.609	0.215	0.617
FE	Month	Month	Month	Year-Month	Year-Month	Year-Month

Notes. The Table provides the impact of an additional daily GWh of wind and solar (in GWh) on wholesale energy prices, AS operating costs, and final prices in €/MWh using equation (1). Prices are weighted by hourly demand to be a better measure of total costs paid by consumers. Renewable generation for wind, solar, and hydro is instrumented with radiation capacity, wind speed capacity, and hydro producible. Demand is instrumented by average daily HDD and CDD, and day-of-week dummies. Fixed Effects (FE) are either at the monthly or year-month level. Standard errors are clustered at the year-month level.

Substitution patterns are very different, need for new solutions





The need for adaptive regulation

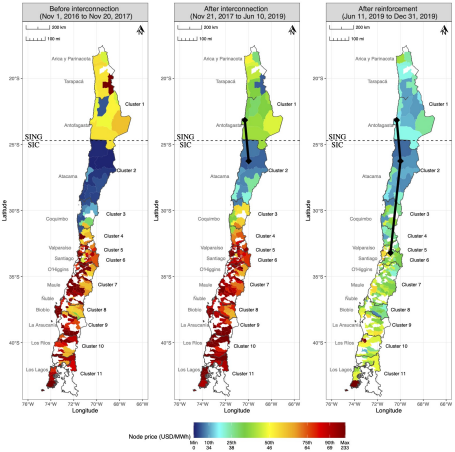
- The Spanish market is working to improve the participation of other units in these reliability markets.
- Some previous experience allowing wind farms to provide 'down' services suggests that competition can increase substantially, and some minor evidence in 2022.
- Properly pricing these services and the dispatch of resources can avoid crowding out better alternatives and should encourage the entry of batteries and more solar in the system.
- Still work to be done: how to improve the functioning of secondary markets?

To be continued...

II. Case study from Chile: Transmission

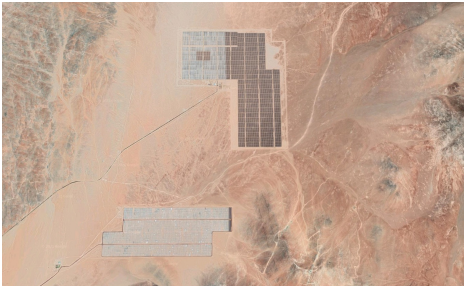
Gonzales, Ito, and Reguant (2023)

- **Question**: What is the cost-benefit of the Chilean ttransmission expansion project?
- **Methods**: event study + structural model of the Chilean electricity market.
- **Co-authors**: Luis Gonzales and Koichiro Ito
- **Findings**:
  - We highlight the dynamic benefits of grid expansion, enabling increased renewable expansion.
  - The cost of transmission can be quickly recovered, even when ignoring the added climate change benefits.

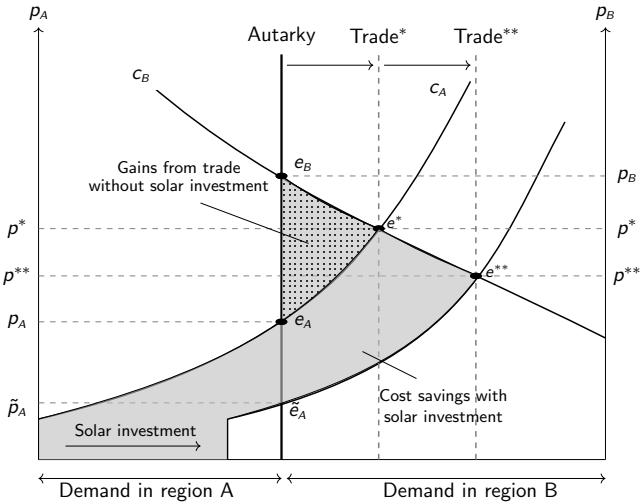


A case study from Chile

- The Chilean context provides a unique case study.
- Chile has large solar resources, but best spots disconnected from demand centers (Antofagasta and Atacama desert).
- Chile successfully connected these areas via ambitious grid projects in 2017 and 2019.
- We provide a *dynamic* quantification of the benefits.



Summary of the paper in a picture



Static impacts: Event study effects of the line

$$c_t = \alpha_1 I_t + \alpha_2 R_t + \alpha_3 c_t^* + \alpha_4 X_t + \theta_m + u_t$$

- Our method uses insights from Cicala (2022)
  - c_t is the observed cost
  - c_t^* is the nationwide merit-order cost (least-possible dispatch cost under full trade in Chile)
  - $I_t = 1$ after the interconnection; $R_t = 1$ after the reinforcement
  - X_t is a set of control variables; θ_t is month fixed effects
  - α_1 and α_2 are the impacts of interconnection and reinforcement

Static impacts: Event study effects of the line

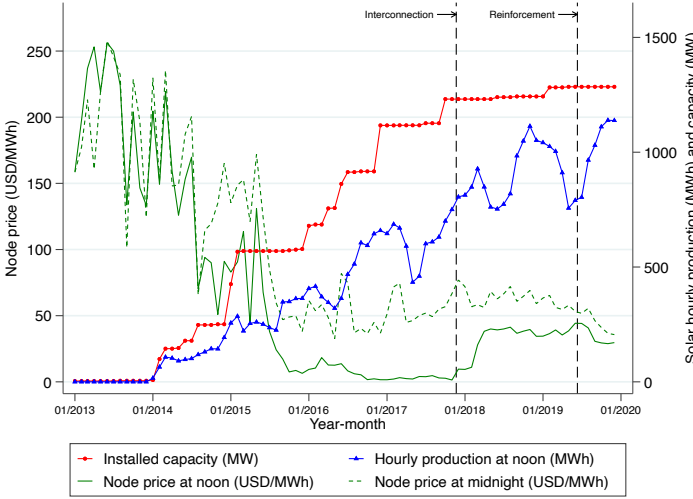
	Hour 12		All hours	
1(After the interconnection)	-2.42	(0.26)	-2.07	(0.17)
1(After the reinforcement)	-0.96	(0.58)	-0.61	(0.37)
Nationwide merit-order cost	1.12	(0.03)	1.03	(0.01)
Coal price [USD/ton]	-0.03	(0.01)	-0.01	(0.01)
Natural gas price [USD/m^3]	-10.36	(4.33)	-0.65	(3.09)
Hydro availability	0.43	(0.14)	0.00	(0.00)
Scheduled demand (GWh)	-0.51	(0.13)	-0.01	(0.00)
Sum of effects	-3.38		-2.68	
Mean of dependent variable	35.44		38.63	
Month FE	Yes		Yes	
Sample size	1033		1033	
R^2	0.94		0.97	

Does this static event study analysis get the full impact?

- Our theory suggested:
  - Yes if solar investment occurs simultaneously with integration
  - No if solar investment occurs in anticipation of integration

Solar investment occurred in anticipation of integration

- Solar investment began after the announcement of integration in 2014
- Plants entered "too early".
  - [→] Static analysis does not capture the full impact of market integration
  - [→] We address this challenge in the next section



Theoretical toy model to inform the "bias"

- With anticipated investment (empirically-relevant case):
  - **Result 1** Static event study analysis understates gross cost savings
  - **Result 2** Static event study analysis understates price reductions
  - **Result 3** Static event study analysis overstates price convergence
- We use both descriptive analysis and structural estimation to:
  - Estimate the full effect of market integration
  - Quantify the impact with and with investment effects

Building a model to get at the full effect

- Impacts of the grid can be static and dynamic:
  - Production benefits: more solar can be sent to the demand centers, prices in solar regions go up.
  - Investment benefits: more solar power is built.

- We highlight that an event study is likely to capture only the first kind of effects (e.g., around time of expansion).
- We build a model of the Chilean electricity market to quantify the benefits of market integration including its investment effects.

A structural model to study a dynamic effect on investment

- We divide the Chilean market to five regional markets with interconnections between regions (now expanding to 11)
- Model solves constrained optimization to find optimal dispatch that minimizes generation cost
- Constraints:
  1. Hourly demand = (hourly supply - transmission loss)
  2. Supply function is based on plant-level hourly cost data
  3. Demand is based on node-level hourly demand data
  4. Transmission capacity between regions:
     - Actual transmission capacity in each time period
     - Counterfactual: As if Chile did not integrate markets

The cost and benefit of the transmission investments

- Cost of the interconnection and reinforcement
  - \$860 million and \$1,000 million (Raby, 2016; Isa-Interchile, 2022)
- Benefit—we focus on three benefit measures
  - Changes in consumer surplus
  - Changes in net solar revenue (= revenue − investment cost)
  - Changes in environmental externalities

Cost-benefit results

Table: Cost-Benefit Analysis of Transmission Investments

	(1)	(2)
Modelling assumptions		
Investment effect due to lack of integration	No	Yes
Benefits from market integration (million USD/year)		
Savings in consumer cost	176.3	287.6
Savings in generation cost	73.4	218.7
Savings from reduced environmental externality	-161.4	249.4
Increase in solar revenue	110.7	183.5
Costs from market integration (million USD)		
Construction cost of transmission lines	1860	1860
Cost of additional solar investment	0	2522
Years to have benefits exceed costs		
With discount rate = 0	14.8	6.1
With discount rate = 5.83%	> 25	7.2
With discount rate = 10%	> 25	8.4
Internal rate of return		
Lifespan of transmission lines = 50 years	6.95%	19.67%
Lifespan of transmission lines = 100 years	7.23%	19.67%

Assessing the cost-benefit

- With the model, we can compute the benefits of the line, with and without investment effects.
- We find that investment effects are key to justify the cost of the line.
- The line was also very attractive from a consumer welfare perspective, even at 5.83% discount rate (Chile's official rate).
- Political economy makes renewable expansion "easy" in Chile.
- How to reduce political economy challenges in other jurisdictions?

In practice, political economy can indeed complicate calculation

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Sweden open to power cable project if Germany reforms, minister says

Swedish energy minister Ebba Busch prepared to move on Hansa PowerBridge project if Berlin overhauls electricity market

But, she added, "the Swedish government would be prepared to move" on the project if Germany split its internal electricity market into bidding zones that would increase the efficiency of its networks and lower prices.

Such reforms would prevent Germany from attracting as much of Sweden's cheaper electricity, generated largely by hydropower, and avoid costs being pushed up for Swedish consumers. Electricity flows on the networks to wherever the highest price demand is.

ELSEVIER

Energy Policy

Volume 187, April 2024, 114053

What we talk about when we talk about electricity: A thematic analysis of recent political debates on Swedish electricity supply

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Sweden sees red over Germany's energy policy

Stockholm (AFP) – **As one of Europe's biggest suppliers of electricity, Sweden is seeing red over Germany's energy policy, which it says ends up punishing Swedish consumers.**

Issued on: 20/12/2024 - 06:31 | 3 min

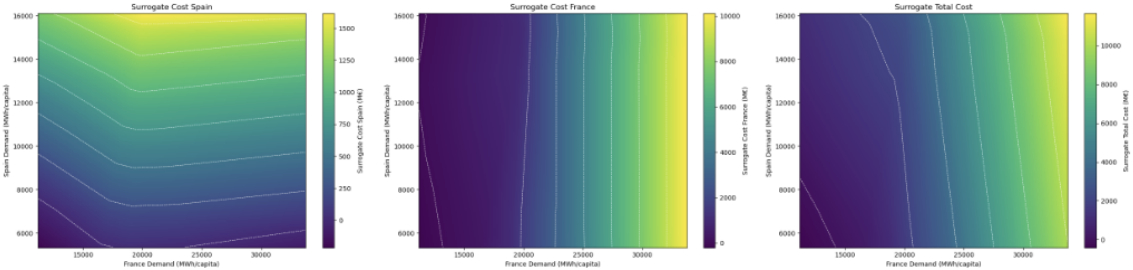
TODAY'S TOP ST

Early work in progress with David Andrés-Cerezo (UAB) and Juan Mercatante (BSE)

- **Question:** How do political interests affect the viability and shape of the network? What is the feasible set and how does it depend on the cost allocation?
- **Methods:** Detailed electricity model of the European network combined with stylized macro models of state members.
- **Computational details:**
  - Use machine learning methods to create a low-dimensional model that accurately reflects the impact of transmission investment on prices and power plant investment.
  - Embed the model into a higher-level macroeconomic model with labor, capital, and electricity as main inputs.

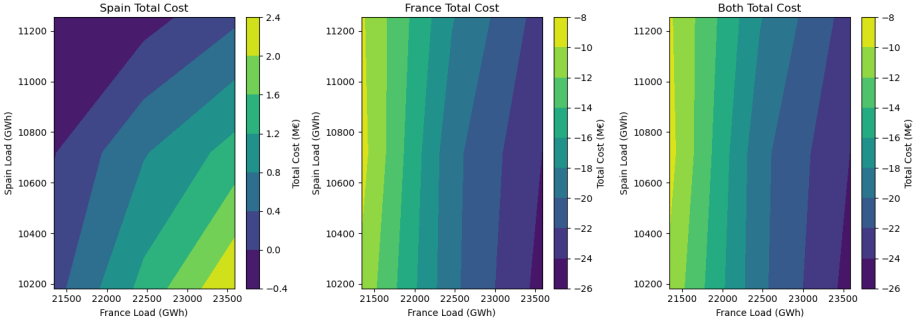
Stylized example with France and Spain

- We solve for the optimal hourly allocation and long-term investment for a given level of aggregate French and Spanish demand (for now, subject to assumptions on hourly behavior by different consumers).
- This approximated surface provides the aggregate optimal costs and the costs to each country, with relationships that can be non-linear.



Stylized example with France and Spain – buidling a line

- We can analyze the benefit of building a larger interconnection between the two.
- Spain wins or loses, but in a minor fashion. France benefits more. Other countries are also affected due to network externalities!
- This contrasts with earlier quantifications of the benefits of the line.
- In addition, it is yet shy of quantifying downstream demand effects (e.g., industrial demand).



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TSOs' estimate

The savings in fuel costs enabled by the interconnector, coined as socio-economic welfare (SEW) in the TYNDP, represent the main part of these benefits. The TYNDP published in 2016 provides the gross estimates for these savings, which have been split between countries by the TSOs in their investment request. The figures are as follows:

Spot year	2020 EP	2030 V1	2030 V2	2030 V3	2030 V4
SEW Europe (M€/yr)	200	120	150	120	240
SEW France (M€/yr)	51	37	19	35	89
SEW Spain (M€/yr)	110	97	162	70	170

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The project was estimated at €1,750m in 2017, half of which would be financed by each partner, with potential cost increases financed by the Spanish transmission system operator (TSO) REE (62.5%) and its French counterpart RTE (37.5%). However, due to unfavourable market context and surging component prices (cables and conversion substations), the estimated total cost has soared to €2,850m, with a risk envelope of €250m. Partners have decided to invest €1,195m each in the project; additional costs between €2,390m and €2,700m will be supported by REE (62.5%) and RTE (37.5%) and extra costs above €2,700m will be financed on a 50:50 basis. Despite rising costs, the project is still considered as beneficial for the two countries and for Europe and it should receive a €578m support from the Connecting Europe Facility (CEF), of which €350m will be granted to RTE.

The need for incorporating industrial policy in the political economy

- Cost allocation is stylized at best, and a source of active discussion.
- The benefits from the transmission line are also a moving target, as new renewable projects are adopted and latent demand shifts (e.g., cooling/heating, artificial intelligence), creating potential commitment problems given the long.
- Need to bring in detailed network analysis with more aggregate macroeconomic forces.

To be continued...

Conclusion

Evaluating the energy transition

- The energy transition provides a unique opportunity to decarbonize electricity generation.
- I evaluated the impacts and challenges of the transition using a diverse set of tools.
- Challenges and concerns remain, lots of areas for economic research.

- **More details?**
  - Measuring the Impact of Wind Power and Intermittency, with Claire Petersen and Lola Segura, *Energy Economics*.
  - The Price Impacts of Renewable Power: A Tale of Two Sources, with David Brown, *IJIO*.
  - The Investment Effects of Market Integration: Evidence from Renewable Energy Expansion in Chile, with Luis Gonzales and Koichiro Ito, *Econometrica*, 91(5): 1659-1693, 2023.

Moltes gràcies!

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